

Chapter 1

Machine Learning-Based Intrusion Detection Systems: Capabilities, Methodologies, and Open Research Challenges

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Abstract: Intrusion Detection Systems (IDSs) are vital for protecting computer networks against unauthorized access and evolving cyber threats. Traditional signature-based IDSs, while effective for known attacks, struggle to identify novel and sophisticated threats. Machine Learning-based Intrusion Detection Systems (ML-IDSs) have emerged as a promising solution, offering enhanced capabilities for threat detection, adaptability, and response. This chapter explores the promising capabilities of ML-IDSs in cybersecurity, focusing on five key strengths. First, machine learning (ML)

algorithms enable the detection of zero-day attacks by identifying anomalous patterns in network traffic without the need for predefined signatures. Second, explainable AI (XAI) integrated with ML models enhances understanding and trust by elucidating the decision-making process behind alerts. Third, ML methods facilitate the detection of intrusions in encrypted traffic, maintaining data privacy while ensuring security. Fourth, the use of Graph Neural Networks (GNNs) introduces context-aware and provenance-based detection, providing structured analysis of relationships within network data. Finally, Large Language Models (LLMs) enhance IDSs by understanding network traffic as human language, detecting complex intrusions, and processing sequential network data for intrusion detection. This paper highlights these capabilities and presents an organized analysis of the state-of-the-art contributions in each area, followed by a detailed discussion of open research challenges and future directions in this important domain. This comprehensive review underscores the transformative role of ML in IDSs in addressing emerging and increasingly powerful cyber threats.

Keywords: Intrusion Detection Systems, Machine Learning, Zero-Day Attack, Anomaly Detection, Explainable AI, Encrypted Traffic Analysis, Advanced Persistent Threats, Provenance-Based Detection, Graph Neural Networks, Large Language Models.

1.1. Introduction

IDSs are essential for safeguarding digital assets by monitoring network traffic and detecting potential threats. As cyber threats become increasingly sophisticated and network infrastructures grow in complexity, coupled with an ever-increasing volume of data, the need for effective IDSs is more critical than ever. Modern IDSs provide automated, scalable solutions that can analyze vast amounts of data, identify anomalies, and offer early warnings to prevent data

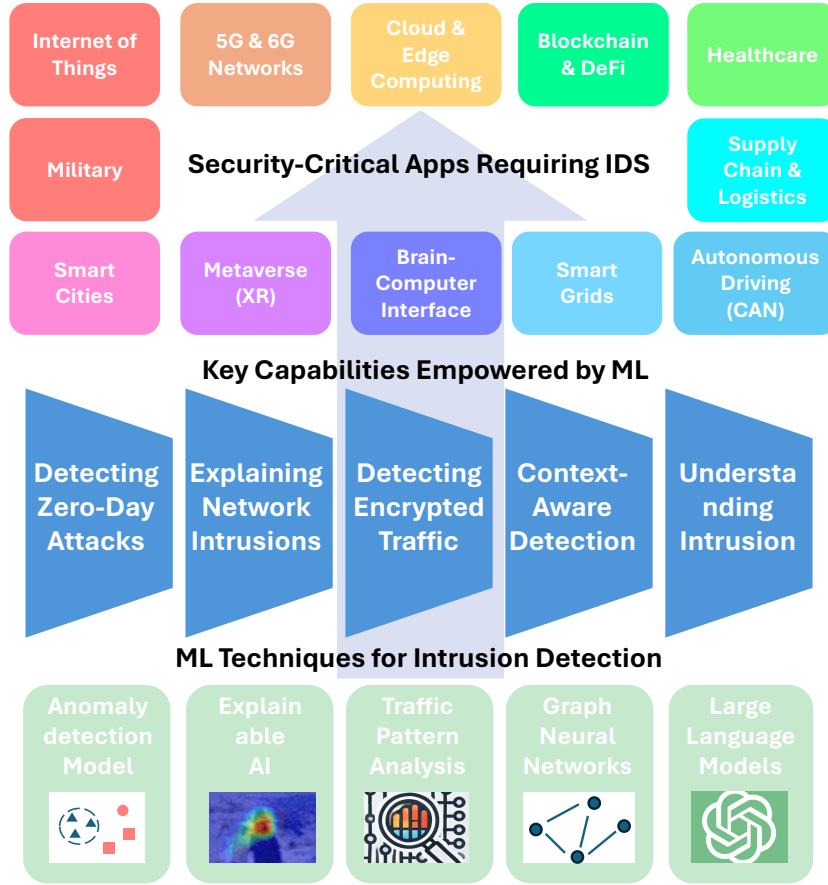


Figure 1.1: Overview of ML-IDS: Core Technologies, Key Capabilities, and Critical Applications.

loss and ensure system integrity. Their role is vital to maintaining robust cybersecurity.

Traditional IDSs rely heavily on signature-based and rule-based detection methods, which, while effective for known threats, often fall short against new, unknown attacks. ML brings transformative potential to IDSs by enabling systems to learn from data, recognize complex patterns, and adapt to evolving threats. ML-IDSs can identify anomalies indicative of cyberattacks, even those without predefined signatures. Beyond this ability, this chapter emphasizes the

promising capabilities that ML facilitates in IDSs.

This chapter presents a vision for ML-IDSs, as shown in Fig. 1.1, identifying core ML technologies such as anomaly detection models, XAI, traffic pattern analysis, GNNs, and LLMs. These technologies empower IDSs with essential capabilities tailored to meet the demands of security-critical applications such as autonomous driving, the Internet of Things (IoT), 5G & 6G networks, cloud & edge computing, blockchain & decentralized finance (DeFi), healthcare, military, supply chain & logistics, the metaverse, brain-computer interfaces, smart grids, and smart cities. This chapter delineates the state-of-the-art contributions of each capability of ML-IDSs and provides future directions for addressing open research challenges in this evolving field.

We summarize the **capabilities** offered by ML (see below and Fig. 1.1):

- **Detecting Zero-Day Attacks:** One of the most compelling benefits of ML in IDSs is its potential to detect zero-day threats. By learning from existing patterns and behaviors, ML algorithms can identify anomalies indicative of novel attacks without relying on prior signatures.
- **Explaining Network Intrusions:** ML models can incorporate explainability frameworks that provide insights into their decision-making processes. This helps cybersecurity analysts understand the rationale behind intrusion alerts, enhancing trust and facilitating informed responses.
- **Detecting Encrypted Traffic:** ML approaches can analyze patterns in encrypted traffic flows to detect potential intrusions without needing to decrypt the data, thus preserving privacy while ensuring security.
- **Context-Aware Threat Detection Using GNNs:** GNNs enable context-aware and provenance-based threat detection. This allows for a more structured understanding of relationships between behaviors in a network, im-

proving the detection of APTs.

- **Understanding Intrusions Through LLMs:** LLMs in IDSs offer the capability to detect complex intrusion patterns by interpreting and processing network data as natural language, fostering understanding of advanced threats and enabling more proactive defenses.

The remainder of this paper is organized as follows: Sec. 1.2 discusses the basic concepts of IDSs and the integration of ML for intrusion detection. Sec. 1.3 covers zero-day attack detection through ML. Sec. 1.4 explores offering interpretability for ML-IDSs. Sec. 1.5 addresses intrusion detection in encrypted traffic using ML. Sec. 1.6 focuses on context-aware and provenance-based threat detection with GNNs. Sec. 1.7 discusses leveraging LLMs for intrusion detection. Finally, Sec. 1.8 provides a summary of the chapter.

1.2. Basic Concepts and ML for Intrusion Detection

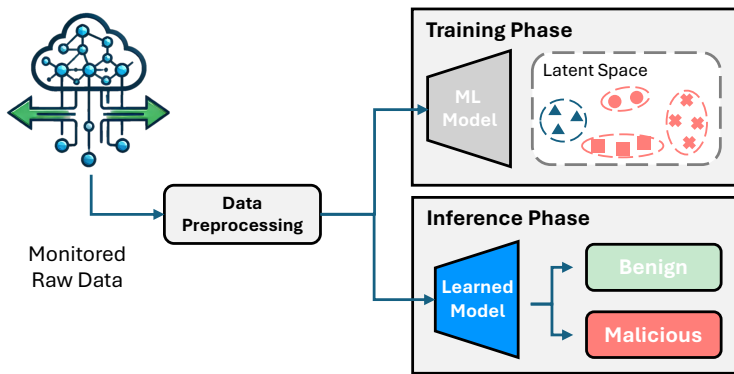


Figure 1.2: A General Workflow of ML-Based IDSs

1.2.1. Fundamental Concepts

IDSs play a pivotal role in safeguarding security-critical applications across a wide spectrum of domains. For instance, autonomous driving systems must address risks such as adversarial attacks on sensors Shahriar et al. [2024a, 2023b,a], while IoT networks face vulnerabilities stemming from constrained devices and their exposure to the physical environment Antonakakis et al. [2017], Shi et al. [2023b]. Similarly, 5G and 6G networks introduce challenges related to user privacy and a massive increase in connected devices, increasing the attack surface Du et al. [2023a, 2022], Yu et al. [2024]. Cloud and high-performance computing environments are vulnerable to data breaches and insider threats, thus requesting tailored security mechanisms Yu et al. [2021], Zhang et al. [2021], Fan et al. [2019], Zhang [2021], while the blockchain and DeFi systems must deal with smart contract exploits and phishing attacks Li et al. [2023]. Critical sectors such as healthcare and military learning systems demand resilience against targeted attacks Shahriar et al. [2024c], Shi et al. [2023a, 2024], Wang et al. [2024b,c], Zhang et al. [2023], while the metaverse, brain-computer interfaces, and smart cities introduce novel attack vectors through immersive technologies and interconnected infrastructures Zhu et al. [2024], Zhang et al. [2024c], Shahriar et al. [2024b], Zhang [2024]. Effective IDSs must adapt to these diverse and evolving threats to ensure the confidentiality, integrity, and availability of data and services across these cutting-edge domains.

To achieve this, integrating ML into IDSs enhances the detection of malicious behaviors by identifying statistical patterns that differentiate benign from malicious activities. The general ML-based IDS methodology comprises three main steps: feature extraction, training, and inference, as illustrated in

Fig. 1.2.

Preprocessing: Network traffic is complex and voluminous, requiring pre-processing to identify and format key features that distinguish benign from malicious activity. This step involves transforming raw network data into a format suitable for ML model input. **Training:** In this phase, labeled or unlabeled data trains the model using either supervised or unsupervised learning to differentiate normal from abnormal behavior. The model maps processed network traffic into a latent space, forming distinct clusters for detecting anomalies. Different ML algorithms can be applied based on IDS requirements, as discussed in Sec. 1.2.2. **Inference:** Once trained, the model is deployed for real-time network monitoring. During inference, preprocessed network traffic is analyzed by the model, enabling the detection of potential threats by comparing live data with learned patterns, thereby enhancing system security.

1.2.2. ML Algorithms for Intrusion Detection

Machine learning algorithms in IDSs are generally categorized into two main types: unsupervised and supervised learning.

Unsupervised Learning Algorithms: Unsupervised learning identifies patterns within data without labels. In IDSs, these algorithms analyze processed network traffic to establish a baseline for normal behavior, flagging deviations as potential intrusions. This is particularly useful for detecting unknown or novel attack types.

Supervised Learning Algorithms: Supervised learning relies on labeled data to train models that classify traffic as benign or malicious. By learning relationships between features and labels, these models can directly detect in-

trusions, clustering benign and attack types distinctly for more accurate threat identification.

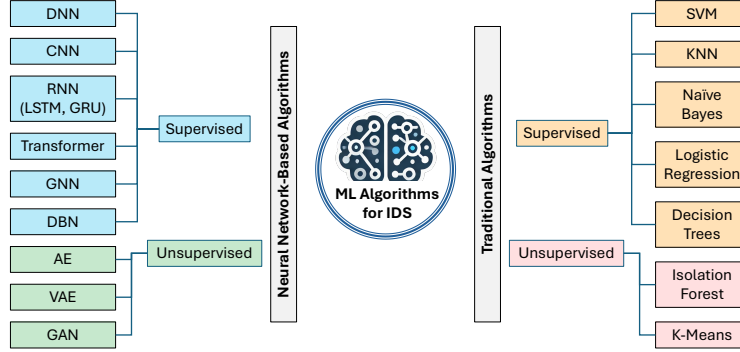


Figure 1.3: Taxonomy of Machine Learning Algorithms for IDSs

A variety of ML algorithms, ranging from traditional methods to advanced neural networks, have been applied to IDSs, as illustrated in Fig. 1.3 and summarized in Ahmad et al. [2021], Liu and Lang [2019].

Traditional ML Algorithms: Traditional ML algorithms have laid the foundation for IDSs by leveraging statistical and pattern recognition techniques. Support Vector Machines (SVM) Cortes [1995] identifies an optimal hyperplane to distinguish between benign and malicious traffic. K-Nearest Neighbor (KNN) Cover and Hart [1967] classifies data points based on proximity to known instances, making them effective for anomaly detection through feature similarity. Naïve Bayes John and Langley [2013] applies Bayes’ theorem with strong independence assumptions to estimate the likelihood of an attack, enabling traffic classification. Logistic Regression Cox [1958] models the probability of malicious behavior using a sigmoid function to predict binary outcomes. Isolation Forest Liu et al. [2008] detects anomalies by isolating outliers through recursive partitioning, making it effective for identifying po-

tential threats. Decision Tree Quinlan [1986] classifies traffic by learning rules that partition data based on feature values, while K-Means Clustering Hartigan and Wong [1979] performs unsupervised grouping of normal and anomalous activities based on feature similarity. These computationally efficient methods have proven effective for basic intrusion detection.

Advanced ML Algorithms with Neural Networks: The advent of neural networks has enabled IDSs to address the complexity of modern attacks with more advanced techniques. Deep Neural Networks (DNNs) LeCun et al. [2015] utilize multiple hidden layers to capture non-linear relationships, making them suitable for recognizing complex attack patterns. Convolutional Neural Networks (CNNs) LeCun et al. [1998] extract spatial features from network data through convolution operations, enabling localized anomaly detection. Recurrent Neural Networks (RNNs) Elman [1990], including Long Short-Term Memory (LSTM) Hochreiter [1997], Sherstinsky [2020] and Gated Recurrent Units (GRUs) Cho et al. [2014], capture temporal dependencies in network sequences, making them effective for attacks with temporal characteristics. Autoencoders (AEs) Hinton and Salakhutdinov [2006] reconstruct input data to identify anomalies based on reconstruction error, while Variational Autoencoders (VAEs) Kingma and Welling [2014] extend this approach probabilistically for improved robustness. Generative Adversarial Networks (GANs) Goodfellow et al. [2014] enhance training datasets by generating synthetic attack samples, improving IDS performance. Transformers model long-range dependencies using self-attention, making them effective for sophisticated attack detection. Graph Neural Networks (GNNs) Kipf and Welling [2016] capture relationships between entities in network data, excelling in detecting multi-node attacks. Deep Belief Networks (DBNs) Hinton et al. [2006] leverage layered

representations to identify intricate patterns in traffic. These advanced algorithms enhance feature extraction, anomaly detection, and scalability, enabling IDSs to counter complex and evolving attack vectors.

1.2.3. Taxonomy of IDSs

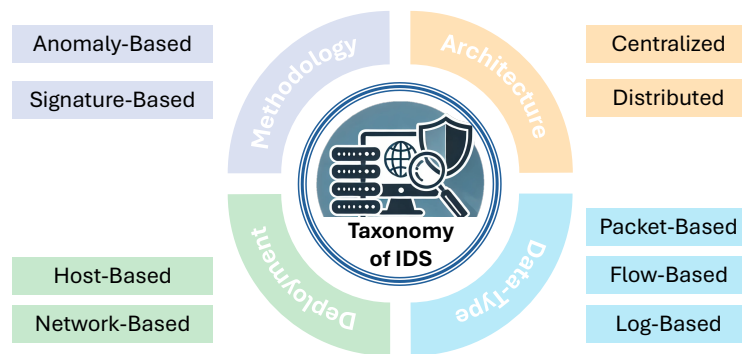


Figure 1.4: Taxonomy of IDSs from the Perspectives of Detection Methodology, Deployment Location, Detection Data Type, and Design Architecture

IDSs can be categorized based on detection methodology, deployment location, data type, and architecture (Fig. 1.4).

Detection Methodology: IDS methods are broadly classified as Signature-Based or Anomaly-Based. A signature-based IDS detects known attack patterns with high accuracy but fails to identify novel threats. An anomaly-based IDS detects deviations from normal activity, enabling novel attack detection but often incurring high false positive rates.

Deployment Location: A host-based IDS monitors specific devices, analyzing system logs and file changes. A network-based IDS monitors network traffic, detecting network-wide attacks.

Detection Data Type: Packet, Flow, and Log-based IDSs provide vary-

ing insights. Packet-based IDSs examine individual packets, which is useful for application-layer threats but resource-intensive. Flow-based IDSs analyze aggregated network flows, enabling scalability and efficient detection of large-scale attacks like Distributed Denial-of-Service (DDoS), though lacking detailed packet-level analysis. Log-based IDSs monitor system events to detect unauthorized activities, useful for identifying misuse and policy violations.

Architecture: Centralized IDSs offer efficient monitoring, though privacy concerns and scalability issues may arise. Distributed IDSs operate locally across devices, fitting decentralized networks like IoT and enhancing privacy by limiting data sharing. Federated Learning enables collaborative model training across nodes, though heterogeneous devices and data present challenges to accuracy and model generalization Wang et al. [2022], Zhang et al. [2023].

1.2.4. Evaluation Metrics and Datasets

Table 1.1: Summary of Widely Used IDS Datasets.

Dataset	Attack Types	Data Type
DARPA IDS	DoS, Probe, R2L, U2R	Packets
KDD Cup'99	DoS, Probe, R2L, U2R	Flow
NSL-KDD	DoS, Probe, R2L, U2R	Flow
N-Balot	Mirai, Bashlite	Flow
CICIoV2024	DoS, Spoofing in IoV	Flow
CICEV2023	DDoS	Flow
ISCX-IDS-2012	Injection, HTTP flooding, DDoS, brute force	Packets & Flow
UNSW-NB15	Backdoors, DoS, Exploits, Fuzzers, Generic, Port scans, Reconnaissance, Shellcode, worms	Packets & Flow
CIC-IDS2017	Brute Force, HeartBleed, Botnet, DoS, DDoS, Web, Infiltration	Packets & Flow
CSE-CIC-IDS2018	HeartBleed, DoS, Botnet, DDoS, Brute Force, Infiltration, Web	Packets & Flow
ADFA-LD	Password Brute Force, R2L, Privilege Escalation, Data Exfiltration	Logs
CERT-Insider-Threat	IT Sabotage, Data Theft Exfiltration, Fraud	Logs
VAST2011-Mini-Challenge2	Insider Threat	Logs

IDS performance is evaluated using metrics derived from the Confusion

Matrix, which summarizes classification outcomes as True Positives (TP, correctly detected attacks), True Negatives (TN, correctly identified normal traffic), False Positives (FP, benign traffic incorrectly flagged as an attack), and False Negatives (FN, undetected attacks). These values enable the calculation of key metrics for assessing IDS effectiveness.

Accuracy: Measures overall correct predictions as:

$$A = \frac{TP + TN}{TP + TN + FP + FN} \quad (1.1)$$

While useful, accuracy can be misleading in imbalanced datasets.

Recall (Sensitivity): Proportion of actual attacks correctly detected:

$$R = \frac{TP}{TP + FN} \quad (1.2)$$

Precision: Indicates the reliability of positive predictions:

$$P = \frac{TP}{TP + FP} \quad (1.3)$$

F1 Score: Harmonic mean of Precision and Recall, balancing detection sensitivity and false positives:

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (1.4)$$

False Positive Rate (FPR): Proportion of normal instances misclassified as attacks:

$$FPR = \frac{FP}{FP + TN} \quad (1.5)$$

These metrics collectively provide a comprehensive view of IDS perfor-

mance, emphasizing detection accuracy, reliability, and minimizing false alarms.

To evaluate the performance of IDSs, widely adopted datasets are summarized in Tab. 1.1. By leveraging these datasets, IDS solutions can be rigorously tested to ensure their effectiveness in detecting a wide range of cyber threats across diverse environments.

1.3. Capability I: Zero-Day Attack Detection with ML

1.3.1. Understanding Zero-Day Attacks and Their Impact

Zero-day attacks exploit unknown vulnerabilities, for which no patches exist, posing a significant cybersecurity threat. These attacks can cause extensive damage before detection, bypassing traditional defenses and compromising critical systems. For example, the 2017 WannaCry ransomware attack Mohrle and Patil [2017] exploited a vulnerability in the Windows SMB protocol, spreading globally within hours and disrupting healthcare, finance, and public services. Similarly, the Mirai botnet attack Antonakakis et al. [2017] leveraged zero-day vulnerabilities in IoT devices, creating a botnet to launch DDoS attacks, impacting millions of internet users.

The unpredictable nature of zero-day attacks challenges traditional signature-based defenses, which rely on known patterns and fail to detect novel exploits. This limitation highlights the need for advanced methods, such as ML-IDSs. By learning normal network behavior, ML-IDSs can identify deviations that

may indicate zero-day attacks. For instance, unusual patterns like the rapid spread of ransomware or unexpected IoT activity can be flagged for investigation, potentially mitigating attack impact.

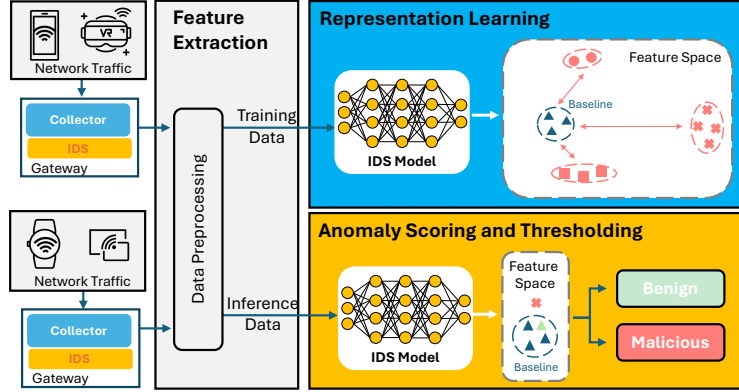


Figure 1.5: Anomaly detection workflow: feature extraction, baseline construction through representation learning, and scoring of new data to identify anomalies as benign or malicious.

1.3.2. General Workflow of ML-IDS for Identifying Zero-Day Attacks

As illustrated in Fig. 1.5, the workflow for detecting zero-day attacks begins with capturing and preprocessing network traffic to extract key features. These features establish a baseline of normal behavior through representation learning. Incoming traffic is then compared to this baseline, with an anomaly score assigned to indicate deviations. A threshold is applied to these scores to classify traffic as benign or malicious, enabling the detection of unknown threats.

Feature Extraction: Anomaly detection starts with extracting key features from network traffic, such as packet sizes, durations, and protocols. These

features capture the statistical properties of the traffic and form the foundation for learning normal patterns. Traditional approaches use predefined statistical features, whereas modern ML techniques enable automated, complex feature extraction directly from raw traffic data.

Representation Learning: Effective anomaly detection depends on accurately modeling normal network behavior in the model’s latent space. Different ML techniques establish this baseline, each offering unique strengths for capturing deviations indicative of zero-day attacks. This baseline serves to identify anomalies by reflecting patterns of typical network behavior.

Anomaly Scoring and Thresholding: After modeling normal behavior, new data instances are assigned anomaly scores that quantify their deviation from learned patterns. These scores can be computed using reconstruction error, distance from clusters, or likelihood estimation within a probabilistic framework. Higher scores indicate a greater likelihood of abnormal activity. Setting an optimal threshold is crucial to accurately detect anomalies while minimizing false positives, allowing the intrusion detection system to adapt to dynamic network conditions.

1.3.3. Anomaly Detection Mechanisms

This section reviews various ML-based IDSs that model normal behavior to detect anomalies in network traffic. Each employs unique methods to learn patterns of benign traffic and identify deviations indicative of malicious activity.

Kitsune: Mirsky et al. [2018] uses an ensemble of Autoencoders to reconstruct normal traffic. High reconstruction errors indicate deviations from

learned patterns, signaling anomalies, including IoT-specific threats.

DIoT: Nguyen et al. [2019] employs GRUs, a type of RNN suited for sequence modeling, to predict upcoming network packets. Low prediction probabilities suggest anomalous traffic, effectively capturing time-dependent attack patterns.

ID-CVAE: Lopez-Martin et al. [2017] flags deviations from the learned distribution in latent space as anomalies, enhancing the detection of complex intrusions.

G-IDS: GAN-based IDSs use a generator to mimic normal traffic and a discriminator to identify anomalies. This adversarial process, as seen in systems like Shahriar et al. [2020], detects subtle, sophisticated attacks, including zero-day exploits.

Extreme Learning Machines and Fuzzy C-Means: EMFC Rathore and Park [2018] combines Fuzzy C-Means clustering for handling data uncertainty with ELM for classification, providing robust anomaly detection through a hybrid approach.

IoTDefender: Fan et al. [2020] uses CNNs to capture spatial correlations in traffic features, achieving high accuracy in detecting DDoS and ARP spoofing attacks.

Passban: Eskandari et al. [2020] applies Isolation Forest and Local Outlier Factor for outlier detection, effectively identifying attacks like port scanning in high-dimensional IoT data.

TDTC: Pajouh et al. [2017] combines Naive Bayes with KNN in a two-tier system, reducing false negatives by re-evaluating instances that may be misclassified.

RF-SD-IoT: Random Forest-based IDSs Zhang et al. [2020], employ en-

sembles of decision trees to enhance robustness and accuracy in detecting abnormal traffic patterns.

Sentinel: It is a kernel-level intrusion detection system for IoT devices that leverages low-level system information to detect malicious events with high accuracy while maintaining minimal power consumption, ensuring compatibility with resource-constrained IoT platforms Cosson et al. [2021].

CIDS: Yue et al. [2022] applies contrastive learning to refine feature representations for enhanced anomaly detection in cybersecurity.

FeCo: Wang et al. [2022] introduces a federated contrastive learning framework tailored for IoT networks, enhancing the representation of normal behavior across distributed devices while preserving privacy by avoiding raw data sharing.

AOC-IDS: Zhang et al. [2024b] enhances traditional Autoencoders by integrating contrastive loss, creating a clearer baseline for normal behavior.

Hermes: Hermes Zhang et al. [2024a] utilizes advanced geometric feature learning Xu and Zheng [2024] to refine latent space representations for anomaly detection. By integrating contrastive learning and H-Score feature learning Xu and Huang [2020], Hermes maximizes the compactness of baseline clusters, enhancing the separation between normal and anomalous instances to improve detection performance.

We summarize the anomaly detection mechanisms in Tab. 1.2.

1.3.4. Open Research Challenges

Despite advances in ML for anomaly detection, key challenges remain:

- **Adapting to Normality Shifts:** Models often struggle with changes in

Table 1.2: Review of ML-Based Anomaly Detection Methods.

Method	ML Mechanism	Dataset	Security Threats
Kitsune (2018)	An ensemble of Autoencoder	Self-generated	Recon., Man-in-the-Middle, DoS, and Mirai Botnet.
DIoT (2019)	Gated Recurrent Units	Self-generated	Mirai Botnet
ID-CVAE (2017), ICVAE (2019)	Conditional variational autoencoder	NSL-KDD, UNSW-NB15	Exploits, Fuzzers, Recon., Analysis, Backdoor, Shellcode, Worms, etc.
G-IDS (2020)	GAN-based learning	NSL-KDD	DoS, Probing, U2R, R2L
EMFC (2018)	ELM-based semi-supervised Fuzzy C-Means	NSL-KDD	DoS, Probing, U2R, R2L
IoTDefender (2020)	Convolutional neural network	NSL-KDD, CICIDS2017	PortScan, DDoS, ARP Spoofing, ARP MitM, Fuzzing, Mirai, etc.
Passban (2020)	Isolation Forest, Local outlier factor	Self-generated	Port Scan, HTTP & SSH Brute Force, SYN flood
TDTC (2019)	Naive Bayes and K Nearest Neighbor	NSL-KDD	DoS, Probing, U2R, R2L
RF-SD-IoT (2020)	Random Forest	Self-generated	SYN, ping flood, UDP port scan, and UDP flood
SENTINEL (2021)	Naive Bayes, Decision Tree, Logistic Regression, and Random Forest Classifier	Self-generated	Network scan, Exfiltration, C&C keep alive, Black/grey hole
CIDS (2019)	Contrastive tasks based on random masking of network packet sequences	NSL-KDD, UNSW NB15	Exploits, Fuzzers, Recon., Analysis, Backdoor, Shellcode, Worms, etc.
FeCo (2022)	Contrastive-learning-based anomaly detection	BaIoT dataset, NSL-KDD	DoS, Probing, U2R, R2L, Mirai, and BashLite
AOC-IDS (2023)	Autoencoder with Contrastive Loss	NSL-KDD, UNSW-NB15	Exploits, Fuzzers, Recon., Analysis, Backdoor, Shellcode, Worms, etc.
Hermes (2024)	Contrastive-learning-based anomaly detection	BaIoT dataset, NSL-KDD	DoS, Probing, U2R, R2L, Mirai, and BashLite

”normal” behavior caused by user, system, or environmental variations. Effective approaches are needed to detect and adapt to these shifts while preserving previously learned patterns.

- **Reducing False Positives:** High false-positive rates burden security teams and erode trust. Advanced clustering techniques and refined feature learning are needed to reduce false positives without compromising accuracy.
- **Privacy-Preserving Collaborative Learning:** Distributed environments require privacy-preserving methods for collaborative data sharing. Federated learning offers promise but faces challenges with biased data distributions.

- **Efficient Model Adaptation with Limited Labeling:** Reducing reliance on extensive labeled data and enabling selective adaptation remain open challenges.
- **Improving Interpretability of Outputs:** XAI techniques are needed to provide actionable insights, helping teams validate and respond to flagged anomalies effectively.
- **Intrusion Classification for Targeted Response:** Accurately classifying intrusions enables more effective response strategies, particularly for distinguishing zero-day attacks from known threats.
- **Scalability for IoT and Edge Devices:** IoT and edge devices require lightweight models that maintain accuracy under resource constraints.

Addressing these challenges is critical for ensuring robust IDS performance against increasingly sophisticated cyber threats.

1.4. Capability II: Intrusion Explainability through XAI

1.4.1. Enhancing Transparency and Trust in Intrusion Detection

With AI and ML playing critical roles in cybersecurity, enhancing IDS explainability is essential. Traditional AI-driven IDSs often operate as "black boxes," providing predictions without transparency, which hampers analysts' ability to understand and respond effectively to alerts. XAI mitigates this issue by

clarifying why certain behaviors are flagged as malicious, benign, or indicative of specific intrusion types.

Explainability is particularly crucial in high-stakes environments like Cyber Security Operations Centers Zimmerman [2014], where analysts rely on IDS alerts to prioritize actions. DARPA’s XAI program Gunning and Aha [2019] underscores the importance of explainable ML for security, aiming to create models that not only detect threats but also justify their decisions. This transparency aids analysts in identifying potential threats, including zero-day exploits, enabling more targeted responses.

Explainable IDSs also reduce false positives, improving transparency and user trust. Clear explanations help analysts differentiate genuine threats from benign anomalies, expediting incident investigations by highlighting the behaviors or features that triggered alerts. This facilitates informed decision-making and a proactive approach to network security, empowering security teams with robust AI tools.

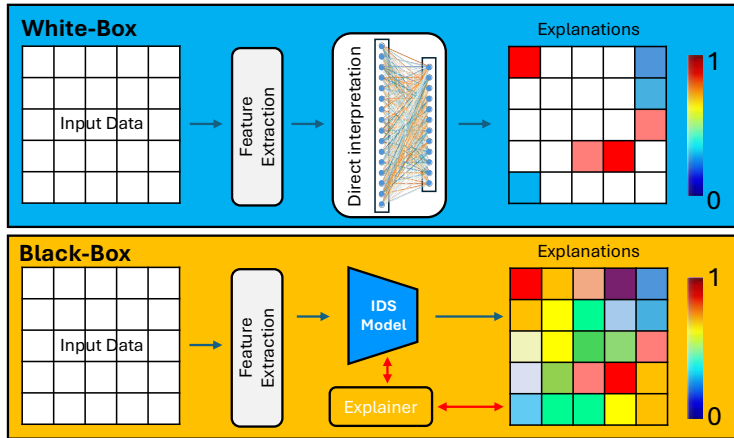


Figure 1.6: Workflow of XAI for ML-IDSs, showing direct interpretability in white-box models and post-hoc explanations for black-box models.

1.4.2. General Workflow of XAI

The XAI workflow for IDS involves several stages. First, the model training phase develops a traffic detection or classification model. Following training, the explanation phase provides insights into the model’s decision-making, highlighting influential features. Figure 1.6 illustrates this workflow.

White-box XAI methods offer built-in interpretability, using transparent models like linear decision models, rule-based systems, and generalized additive models Xu et al. [2019], Dwivedi et al. [2023]. These models simplify interpretation by visualizing feature contributions, but their simplicity may limit effectiveness in capturing complex network traffic patterns involving non-linear relationships. The white-box workflow is shown in Fig. 1.6.

In contrast, **black-box** XAI methods provide post-hoc explanations for opaque models such as deep neural networks. Techniques like Local Interpretable Model-Agnostic Explanations (LIME) Ribeiro et al. [2016], SHapley Additive exPlanations (SHAP) Scott et al. [2017], and saliency maps Dabkowski and Gal [2017] analyze model predictions to identify the most influential input features (Fig. 1.6). These methods explain why certain traffic flows are classified as malicious or benign, even for models with limited interpretability.

1.4.3. XAI Methods for IDS Transparency Enhancement

In this work, we discuss several representative XAI methods that can enhance transparency and provide valuable insights into ML-IDSs.

SHAP: Shapley Additive Explanations Scott et al. [2017] use game theory

to assign feature importance, providing both global and local interpretability. By quantifying each feature’s impact, SHAP offers fair and consistent explanations, making it ideal for complex, non-linear models.

LIME: Local Interpretable Model-Agnostic Explanations Ribeiro et al. [2016] generate interpretable local models around each prediction. By perturbing input data, LIME provides instance-specific explanations for any model, aiding in the interpretation of opaque classifiers.

Integrated Gradients: Integrated Gradients Sundararajan et al. [2017] address gradient limitations by integrating gradients from a baseline to the input, offering smooth feature importance and satisfying key axioms for reliable explanations in complex models.

AI2: AI2 Gehr et al. [2018] applies abstract interpretation to approximate neural network behavior across input ranges, enhancing robustness against adversarial attacks. This scalable white-box method ensures reliable performance in security-critical applications.

LEMNA: LEMNA Guo et al. [2018] approximates non-linear models with local linear regressions, tailored for explaining security-focused anomaly detection. By partitioning input space, LEMNA provides clear explanations for complex feature interactions.

Grad-CAM: Grad-CAM Selvaraju et al. [2017] visualizes regions influential in CNN predictions by computing gradients for feature maps in the final convolutional layer. It generates heatmaps that highlight critical input regions, widely used for spatial feature importance in computer vision.

CAM: Class Activation Mapping Zhou et al. [2016] enables CNNs to locate regions relevant to specific class predictions. Relying on global average pooling, CAM produces class-specific activation maps, which are helpful in classification

and weakly supervised localization by identifying regions critical for model predictions without detailed annotations.

Meaningful Perturbation: This method Fong and Vedaldi [2017] identifies the smallest input perturbations that significantly alter predictions, revealing the input’s most critical regions. Unlike gradient-based methods, it operates independently of model architecture, offering a model-agnostic approach for black-box explanations by directly highlighting essential parts of the input.

Real-Time Saliency: A rapid approach for generating saliency maps Dabkowski and Gal [2017], Real-Time Saliency highlights crucial input regions by training a separate network to produce masks. Its efficiency enables interpretability during inference, adapting well to models beyond image classifiers.

QII: Quantitative Input Influence Datta et al. [2016] quantifies feature impact on model decisions by applying a causal framework. By observing output changes due to feature perturbations, QII aids in uncovering potential security vulnerabilities, especially valuable in applications needing transparency on specific feature influences.

LRP: Layer-wise Relevance Propagation Bach et al. [2015] decomposes predictions into relevance scores for input features, making it well-suited for tasks like image classification. By attributing relevance to each pixel, LRP provides clear explanations for non-linear classifiers.

DeepLIFT: Deep Learning Important Features Shrikumar et al. [2017] assigns importance by propagating differences in activations through the network, allowing clear identification of feature contributions, especially useful in deep models where understanding relative feature importance is critical.

CADE: Concept Drift Explanation Yang et al. [2021] explains how model

Table 1.3: Review of XAI Methods for Anomaly Detection.

Method	Key Approach	Model Type
LIME (2016)	Perturbs input data and fits surrogate models locally	Black-box
SHAP (2017)	Uses Shapley values to attribute feature importance	Black-box
QII (2016)	Measures feature influence using causal inference	Black-box
Meaningful Perturbation (2017)	Perturbs input regions to identify critical features	Black-box
Real-Time Saliency (2017)	Generates real-time saliency maps to highlight important regions	Black-box
Integrated Gradients (2017)	Computes gradients to explain feature importance	White-box
Grad-CAM (2017)	Uses gradients to generate heatmaps for CNN explanations	White-box
LRP (2015)	Propagates relevance scores for pixel-wise explanations	White-box
AI2 (2018)	Uses abstract interpretation to certify the neural network by input	White-box
LEMNA (2018)	Approximates non-linear models with local linear models	White-box
DeepLIFT (2017)	Propagates activation differences to attribute importance	White-box
CAM (2016)	Generates class activation maps to highlight important regions	White-box
CADE (2021)	Detects and explains concept drift	White-box
DeepAID (2021)	Interprets and improves anomaly detection models	White-box
XNIDS (2023)	Explains DL-NIDS, generating actionable defense rules for active responses	White-box

predictions change over time by identifying samples affected by drift. This is particularly useful in dynamic environments like security, where models must adapt to evolving threats.

DeepAID: DeepAID Han et al. [2021a] interprets deep anomaly detection models, offering insights into model behavior and strategies to enhance anomaly detection accuracy in real-world security scenarios.

XNIDS: XNIDS Wei et al. [2023] approximates historical input patterns to explain IDS decisions, capturing feature dependencies and generating actionable defense rules. It optimizes for fidelity and adaptability, particularly valuable in real-time network security.

We summarize the XAI methods for IDS and present them in Tab. 1.3.

1.4.4. Open Research Challenges

Despite advancements, several open research challenges limit the effectiveness and applicability of XAI for IDSs in real-world scenarios:

- **Addressing Instability and Uncertainty:** XAI methods often produce feature importance based on estimations, which can be inconsistent across model instances. Improving the stability and reliability of explanations is critical, especially in high-stakes security applications.
- **Adaptability in Dynamic Environments:** Network traffic patterns evolve constantly, requiring ML-IDSs to adapt autonomously. Systems must dynamically adjust to new conditions to maintain effectiveness in changing environments.
- **From Local to Global Explanations:** Most XAI methods focus on local explanations, offering insights for individual decisions but lacking a holistic view. Combining local and global interpretability could help analysts identify trends and vulnerabilities more comprehensively.
- **Robustness Against Adversarial Attacks:** Attackers can exploit interpretability to evade detection. Strategies are needed to enhance ML-IDS resilience, ensuring effectiveness even under adversarial conditions.
- **Tailored Explainability for IDSs:** A standardized definition of explainability specific to IDSs is needed. Developing domain-specific requirements that align with principles like confidentiality, integrity, and availability would enable consistent evaluation and advancement in ML-IDS.
- **User-Centric Explanations:** Different stakeholders, such as security analysts, developers, and managers, have unique needs. Adaptive explanation frameworks should be developed to meet the distinct requirements of each user group, improving the usability and relevance of ML-IDS.

Addressing these research problems will significantly advance ML-IDSs, making systems more transparent, adaptive, and reliable for practical cybersecurity applications.

1.5. Capability III: Intrusion Detection in Encrypted Traffic

1.5.1. Challenges in Intrusion Detection for Encrypted Traffic

While encryption safeguards sensitive information from eavesdroppers, it also limits visibility into traffic content, creating blind spots that attackers can exploit to conceal malicious activities. Effective IDSs for encrypted traffic are essential to detect threats without compromising privacy, ensuring encrypted channels remain secure. Detecting intrusions in encrypted traffic presents significant challenges. Traditional IDS methods, such as deep packet inspection, are ineffective because encryption conceals packet payloads, preventing direct content analysis Papadogiannaki and Ioannidis [2021]. Consequently, IDSs must rely on indirect indicators, such as metadata and flow characteristics like packet size, timing, and frequency patterns. Privacy concerns further complicate IDS for encrypted traffic, as decryption-based solutions risk exposing sensitive data. Developing effective IDSs for encrypted traffic requires balancing security, privacy, and performance—a formidable task given current technical and privacy constraints.

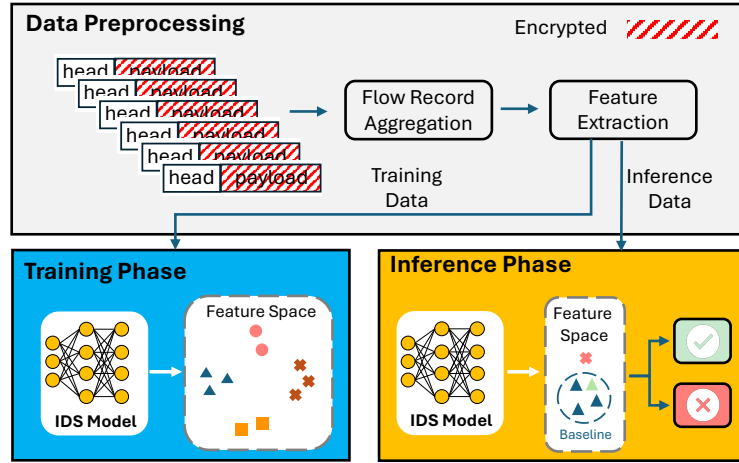


Figure 1.7: General Workflow of IDS for Encrypted Traffic

1.5.2. Workflow of ML-IDS for Encrypted Traffic

ML-IDSs for encrypted traffic focus on analyzing non-content-based features. IDSs analyze metadata such as packet size, timing, frequency, and flow direction to detect abnormal traffic patterns. These patterns are aggregated over time and analyzed using machine learning models trained to classify traffic or identify anomalies. The process begins with preprocessing, where metadata features are extracted and organized. These features are then aggregated for model training, enabling the detection of traffic types or anomalies. During inference, IDSs analyze statistical features to identify threats while preserving user privacy. By focusing on metadata and behavioral patterns, ML-IDSs effectively secure encrypted traffic without compromising its confidentiality (see Fig. 1.7).

1.5.3. ML-Based Solutions for Encrypted Traffic Analysis

In this section, we review several key approaches to detecting encrypted malicious traffic, highlighting their main ideas and methods.

BotFinder: Tegeler et al. [2012] detects bots in network traffic without deep packet inspection, relying solely on high-level features. Using ML to identify command-and-control patterns, BotFinder achieves high detection rates on NetFlow data from large ISPs without analyzing encrypted packet content.

Disclosure: Bilge et al. [2012] identifies botnet command-and-control servers using NetFlow data. By analyzing flow size, access patterns, and temporal behavior, it distinguishes malicious flows in encrypted traffic. Reputation scores further reduce false positives, enabling large-scale real-time detection.

A&M1: Anderson and McGrew [2016] classifies encrypted malware traffic using TLS handshake metadata, DNS flows, and HTTP headers. By leveraging contextual flow data, this approach improves ML classifiers for encrypted traffic.

A&M2: Anderson and McGrew [2017] enhances ML classification for encrypted malware traffic by addressing label inaccuracies and non-stationary data. Feature engineering, guided by domain expertise, and Random Forests proved effective for improving classification performance.

DeepLog: Du et al. [2017] employs LSTM networks to detect anomalies in system logs. Modeling logs as natural language sequences, it identifies deviations in real time, correlating anomalies with encrypted malicious traffic.

RADAR: Zheng et al. [2018] uses software-defined networking for real-time DDoS detection in encrypted traffic. Adaptive correlation analysis identi-

fies attack features, enabling traffic throttling via commercial software-defined networking switches.

Kitsune: Mirsky et al. [2018] introduces a lightweight IDS using an ensemble of autoencoders to detect deviations in encrypted traffic in real time. Kitsune’s resource-efficient design is ideal for deployment on constrained devices like routers and gateways.

FlowLens: Barradas et al. [2021] uses programmable switches to extract flow features at line speed. Compact ”flow markers” classify encrypted traffic efficiently, balancing accuracy and memory constraints for real-time classification tasks.

ET-BERT: Lin et al. [2022] leverages transformer-based models for encrypted traffic classification. Pre-trained on large-scale traffic data, ET-BERT achieves state-of-the-art performance across various tasks, especially on unseen data.

MT-Flowformer: Zhao et al. [2022] employs a semi-supervised framework with an attention-based classifier for encrypted traffic. Using ”Mean Teacher” learning and spatiotemporal augmentation, it achieves superior performance on real-world datasets.

HyperVision: Fu et al. [2023] applies graph learning to detect encrypted malicious traffic. By creating a flow interaction graph, it achieves high detection throughput and outperforms state-of-the-art methods without labeled data.

RAPIER: Qing et al. [2024] tackles label noise in training data for encrypted traffic detection. By correcting noisy labels and augmenting datasets, RAPIER achieves notable F1 score improvements on public and real-world datasets.

Cisco Secure Network Analytics: Cisco [2024] detects threats in en-

Table 1.4: Review of ML-Based Solutions for Encrypted Traffic Analysis

Method	Key Approach	Detection Mechanism
BotFinder (2012)	Detects bots using high-level NetFlow features without packet inspection.	Flow Analysis
Disclosure (2012)	Identifies botnet C&C traffic using flow size, patterns, and reputation scores.	Flow Analysis
Kitsune (2018)	Real-time anomaly detection using ensemble autoencoders on IoT devices.	Flow Analysis
FlowLens (2021)	Classifies encrypted flows using programmable switches and flow markers.	Flow Analysis
A&M1 (2016)	Uses metadata (TLS handshake, DNS, HTTP) to classify malware traffic.	Metadata Analysis
A&M2 (2017)	Applies Random Forests with feature engineering to handle noisy data.	Metadata Analysis
ET-BERT (2022)	Uses transformer models for encrypted traffic classification.	Context Analysis
MT-Flowformer (2022)	Semi-supervised attention-based classifier with data augmentation.	Context Analysis
Cisco Secure Network Analytics (2024)	Detects encrypted threats with analytics and enriched contextual data.	Context Analysis
HyperVision (2023)	Graph learning on flow interaction graphs for unsupervised detection.	Context Analysis
RAPIER (2023)	Corrects label noise and augments datasets for robust traffic classification.	Distribution Analysis
DeepLog (2017)	Detects anomalies in system logs using LSTM-based sequence modeling.	Log Analysis
RADAR (2018)	Detects DDoS attacks in encrypted traffic with adaptive SDN correlation.	SDN switches Analysis

encrypted traffic using advanced analytics without compromising privacy. Contextualized alerts integrate user, device, and application data, enabling actionable insights for malware detection and insider threat mitigation.

We summarize the encrypted traffic analysis methods for IDS and present them in Tab. 1.4.

1.5.4. Open Research Challenges

Intrusion detection in encrypted traffic presents unique open research challenges:

- **Optimizing Detection Performance:** The lack of content visibility in encrypted traffic increases the FPR. Research is needed to improve the recall-

FPR trade-off to enhance detection reliability.

- **Feature Engineering for Encrypted Traffic:** Identifying effective features in encrypted traffic is challenging due to limited payload visibility. Advancing automated feature engineering for encrypted data representations is crucial.
- **Handling Data Imbalance:** Malicious traffic constitutes a small fraction of total network traffic, with encrypted malicious traffic being even harder to collect. Synthetic data generation offers a promising solution for addressing data imbalance.
- **Unsupervised Detection:** Current methods often rely on supervised learning and prior knowledge of attack patterns, limiting their effectiveness against unknown threats. Developing unsupervised learning schemes is essential for detecting novel attacks.
- **Adapting to Evolving Encryption Protocols:** Rapid changes in secure communication protocols and attack methods challenge the adaptability of ML-IDSs. Systems must dynamically learn and adapt to new encrypted traffic patterns to remain effective in dynamic environments.

Addressing these challenges will advance the field of encrypted traffic intrusion detection, enabling systems that are not only accurate and adaptive but also aligned with security and privacy standards.

1.6. Capability IV: Context-Aware Threat Detection and Reasoning with GNNs

1.6.1. Introduction to GNNs in IDS

GNNs have emerged as a powerful tool for intrusion detection, addressing the limitations of traditional ML models that rely on flat data structures like vectors or grids. GNNs operate on graphs, representing data as nodes and edges to capture relationships between entities. A **graph** consists of nodes connected by edges, which is especially well-suited for modeling network data in cybersecurity, where entities such as hosts, processes, or IP addresses interact in complex, interconnected ways.

By applying learning schemes directly to graph structures, GNNs can model the intricate relationships inherent in network interactions. This capability is particularly effective for identifying advanced cyber threats, such as APTs and zero-day attacks, which involve subtle, evolving patterns over time. Traditional models often struggle to capture these non-linear and high-dimensional relationships.

Through graph representation learning, GNNs effectively model both benign and malicious interactions among network entities, making them adaptable to complex cybersecurity scenarios. Their adaptability has enabled success across various applications, including vulnerability detection, threat intelligence, and malware identification, making GNNs a resilient approach for countering sophisticated cyber threats.

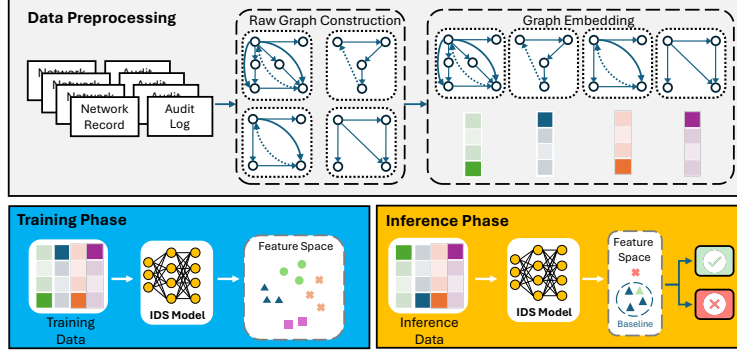


Figure 1.8: General Workflow of GNN-based IDS with Context-Awareness Capability

1.6.2. Workflow of GNNs for Intrusion Detection

GNNs enhance intrusion detection by transforming network data into graphs. This structure captures relational and contextual information critical for detecting multi-stage attacks that evade traditional methods. The workflow of GNN-based IDS, illustrated in Fig. 1.8, involves graph construction, feature embedding, message passing, and anomaly detection.

The process begins with converting raw network data into graph structures tailored to specific needs. **Packet graphs** represent fine-grained, packet-level interactions, where nodes are IP addresses or ports and edges denote packet transfers. Although detailed, packet graphs can pose scalability challenges. **Flow graphs** aggregate packets into flows, summarizing communication patterns to reduce complexity while preserving essential insights. **Authentication graphs** represent users or devices as nodes and authentication attempts as edges, identifying suspicious access patterns. **Host graphs** focus on system-level interactions, such as processes and files, aiding malware detection.

After constructing the graph, GNNs build node and edge embeddings for dimensionality reduction. This enables GNN models to learn patterns for effective differentiation between normal and malicious behavior from graph data during the training phase.

Using these embeddings, GNNs perform detection during the inference phase. For large-scale threats like DDoS, GNNs provide a graph-level overview, enabling comprehensive threat detection. By modeling complex relationships, GNN-based IDS offer robust and scalable solutions for evolving cyber threats.

1.6.3. Provenance-Based Intrusion Detection by GNNs

The following studies represent significant contributions to this field. We summarize these state-of-the-art approaches as follows:

NoDoze: Hassan et al. [2019] focuses on reducing alert fatigue in large enterprises by using provenance data to contextualize threat alerts. It creates causal dependency graphs and assigns anomaly scores based on event frequency, diffusing scores across dependencies to highlight truly suspicious activity.

SLEUTH: Hossain et al. [2017] is a system for reconstructing attack scenarios in real time using a main-memory dependency graph of audit-log data. It employs tag-based detection techniques for attack reconstruction, effectively summarizing large-scale attacks by generating compact visual graphs that assist analysts in identifying attack sequences.

HOLMES: Milajerdi et al. [2019b] detects APTs through the correlation of suspicious information flows. HOLMES produces a detection signal by an-

alyzing coordinated activities within an APT campaign, generating high-level attack graphs that summarize the attacker’s steps, providing a useful tool for real-time response to coordinated threats.

POIROT: Milajerdi et al. [2019a] uses kernel audit logs to align attack behaviors with CTI standards, facilitating threat hunting. It models this process as a graph-matching problem, correlating attack indicators with provenance graph patterns, effectively identifying campaign stages in real time and enhancing investigative efficiency.

RapSheet: Hassan et al. [2020] introduces tactical provenance graphs for endpoint detection and response systems. It reduces log retention overhead by maintaining a skeleton graph, allowing for real-time alert scoring based on alert temporal relationships, thus addressing response systems’ limitations in handling long-term attacks and reducing investigation time.

Morse: Hossain et al. [2020] aims to overcome dependence explosion in forensic analysis. By applying tag attenuation and decay to prioritize significant events, Morse creates compact scenario graphs, filtering out benign background activities while capturing crucial attack information, showing high efficacy in identifying stealthy attacks.

PROVDETECTOR: Wang et al. [2020] targets stealthy malware detection via provenance monitoring. Leveraging neural embeddings, PROVDETECTOR identifies deviations in OS-level behaviors, detecting malicious activities hidden within benign processes, achieving high detection accuracy in enterprise environments.

UNICORN: Han et al. [2020] is an APT detector that uses provenance graph sketching for space-efficient analysis of long-running processes, detecting stealthy attacks without relying on predefined signatures. UNICORN demon-

strates effectiveness in identifying APTs through temporal analysis and space-efficient sketching.

SIGL: Han et al. [2021b] ensures software installation integrity via provenance graph learning. Using graph LSTM and MLP-based autoencoder architectures, SIGL detects malicious installations, highlighting suspicious processes with high accuracy, making it effective against advanced supply-chain attacks.

DEPCOMM: Xu et al. [2022] introduces graph summarization for attack investigation by partitioning large graphs into process-centric communities. It reduces graph size while preserving critical information, effectively identifying attack-related processes and easing analysis for security teams.

SHADEWATCHER: Zeng et al. [2022] utilizes recommendation-inspired techniques to detect cyber threats, predicting entity interactions within audit records. By employing high-order connectivity modeling, it achieves precise threat detection and dynamically adapts to reduce false alarms, pinpointing threats swiftly.

R-CAID: Goyal et al. [2024] integrates root cause analysis in provenance-based IDS. By linking entities to immutable root causes, R-CAID resists adversarial evasion and outperforms baseline graph and log-based IDS in both passive and active adversarial models.

PROGRAPHER: Yang et al. [2023] addresses dependency explosion by embedding provenance snapshots. Using sequence-based learning, PROGRAPHER captures structural graph features and provides key insights to analysts, reducing their workload significantly in anomaly detection.

KAIROS: Cheng et al. employs a GNN encoder-decoder to track structural evolution in provenance graphs for detecting anomalies. It reconstructs attack footprints effectively, generating compact graphs that provide precise

insights for timely response.

MAGIC: Jia et al. [2024] uses masked graph representation learning to detect APTs at multiple granularities, handling concept drift with model adaptation. MAGIC demonstrates high detection performance and low overhead across different datasets.

FLASH: Rehman et al. [2024] leverages GNN-based embeddings for scalable IDS on provenance graphs, employing an embedding recycling database for real-time detection. FLASH outperforms existing models, demonstrating scalability and resilience to mimic attacks in real-world tests.

Table 1.5: Review of Provenance-Based Intrusion Detection Approaches Using GNNs

Method	Key Approach	Purpose
HOLMES (2019)	Correlates suspicious flows to detect coordinated APT campaigns.	APT Detection
PROVDETECTOR (2020)	Detects stealthy malware by identifying deviations in OS-level behaviors.	APT Detection
UNICORN (2020)	Detects APTs using space-efficient provenance graph sketching.	APT Detection
PROGRAPHER (2023)	Reduces dependency explosion with sequence-based learning for anomaly detection.	APT Detection
MAGIC (2024)	Detects APTs using masked graph representation learning with adaptive models.	APT Detection
FLASH (2024)	Uses GNN embeddings with a recycling database for real-time scalable IDS.	APT Detection
SLEUTH (2017)	Reconstructs attack scenarios in real time using dependency graphs from audit logs.	APT Analysis
DEPCOMM (2022)	Summarizes large graphs into communities for efficient attack investigation.	APT Analysis
R-CAID (2024)	Links entities to root causes for resilience against adversarial evasion.	APT Analysis
KAIROS (2024)	Tracks structural evolution in provenance graphs for effective anomaly detection.	APT Analysis
POIROT (2019)	Matches provenance graph patterns with CTI standards for real-time threat hunting.	Forensic Analysis
Morse (2020)	Prioritizes significant events with tag attenuation, filtering benign activities.	Forensic Analysis
SHADEWATCHER (2022)	Predicts entity interactions within audit records for precise threat detection.	Forensic Analysis
NoDoze (2019)	Reduces alert fatigue using causal dependency graphs to assign anomaly scores.	Performance Optimization
RapSheet (2020)	Uses Tactical Provenance Graphs for real-time scoring and reducing log overhead.	Performance Optimization
SIGL (2021)	Ensures installation integrity using graph LSTM and autoencoders.	Installation Integrity

We summarize the ML-IDS using GNNs and present them in Tab. 1.5.

1.6.4. Open Research Challenges

GNN-based IDSs present promising capabilities for detecting complex cyber threats. However, several open research challenges persist in enhancing their applicability:

- **Improving Detection Accuracy and Reducing False Positives:** Balancing detection accuracy with low FPR remains a core issue. Further research is required to refine graph embedding techniques and ensure fine-grained threat detection in large datasets.
- **Scalability of Provenance Graphs:** Provenance-based IDSs face significant challenges in handling large-scale graphs. Processing vast and complex provenance graphs remains computationally intensive and hinders the performance crucial for detecting sophisticated APTs.
- **Temporal and Contextual Information Extraction:** Accurately capturing temporal and causal relationships within provenance graphs is essential. Ensuring both spatial and temporal accuracy under dynamic network conditions is an ongoing research area.
- **Adaptive Detection under Concept Drift:** Continuous system changes introduce concept drift, challenging IDSs to adapt effectively. Designing mechanisms to adjust models dynamically to account for evolving attack patterns without significant increases in false positives is an open problem.
- **Robustness against Evasion and Mimicry Attacks:** Provenance-based IDSs are often vulnerable to evasion techniques, where attackers mimic legitimate processes to avoid detection. Root cause analysis is needed to improve

resilience against such attacks as an open research problem.

By addressing these challenges, future research can further advance the capabilities of graph-based IDSs, enhancing their ability to detect sophisticated cyber threats in diverse and dynamic network environments.

1.7. Capability V: LLMs for Intrusion Detection and Understanding

1.7.1. The Role of LLMs in Cybersecurity

The rise of ML-IDS significantly enhanced the detection of emerging and complex cyber threats. Recent advancements in AI, particularly LLMs, provide transformative capabilities for IDS, improving threat detection, response, and understanding. LLMs like ChatGPT, trained on massive textual corpora, excel in understanding and generating human-like text. While originally developed for natural language tasks, their potential in cybersecurity is profound. We illustrate several security-oriented applications from recent years as follows.

LLMIF Wang et al. [2024a] leverages LLMs to address challenges in IoT fuzzing, automating test case generation for protocols like Zigbee. By extracting and reasoning about unknown message formats, LLMIF improves coverage and uncovers previously hidden vulnerabilities. **CellularLint** Rahman et al. [2024] applies LLMs to 4G/5G security, using domain-adapted few-shot learning to identify inconsistencies in protocol specifications, enhancing privacy and integrity. **CHATAFL** Meng et al. [2024] utilizes LLM reasoning to construct grammar rules and predict stateful protocol message sequences, surpassing

traditional fuzzers in vulnerability detection. **PentestGPT** Deng et al. [2024] automates penetration testing tasks such as tool output interpretation and action recommendations, streamlining workflows and improving task completion rates.

LLMs’ ability to process vast datasets, uncover hidden patterns, and infer complex relationships makes them excel in understanding cyber threats for IDS through several critical aspects:

- **Contextual Understanding:** Using attention mechanisms, LLMs model complex relationships in network traffic and logs, identifying patterns that traditional models may miss, enhancing anomaly detection.
- **Traffic Analysis as Language:** LLMs treat network traffic and logs as structured language, enabling nuanced interpretations that reveal hidden indicators of compromise, significantly boosting detection efficacy.
- **Semantic Understanding:** LLMs semantically interpret commands, logs, and threat vectors, identifying sophisticated attacks, including those with subtle similarities to known vulnerabilities.
- **Adaptation to Evolving Threats:** Fine-tuned on domain-specific datasets, LLMs adapt to emerging and unseen attack vectors, ensuring robust detection against ever-changing threats.

1.7.2. Leveraging LLMs for Intrusion Detection

At the heart of LLMs lies the **attention mechanism**, which dynamically focuses on relevant input sequences, identifying critical relationships across long contexts. This capability makes LLMs exceptionally suited for detecting subtle patterns in cybersecurity data. For detailed surveys on LLMs, see Yao

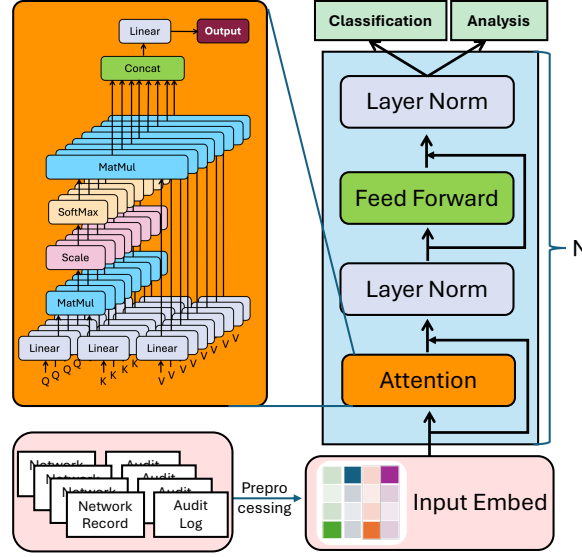


Figure 1.9: Workflow of GPT Models in Intrusion Detection Systems

et al. [2024], Minaee et al. [2024]. The general workflow of LLMs in IDS is outlined below.

Training and Fine-Tuning: Pre-trained LLMs, such as GPT or BERT, are fine-tuned on cybersecurity-specific data, enabling them to capture the unique characteristics of normal and malicious behaviors. This transfer learning approach efficiently adapts the model to new domain knowledge.

Inference and Detection: During inference, LLMs analyze data streams to detect intrusions. Attention mechanisms prioritize critical data points, modeling complex relationships to identify anomalies indicative of threats. As illustrated in Fig. 1.9, LLM-based IDS preprocess data (e.g., logs and traffic records) via tokenization and embedding. Layers of **Attention Mechanisms**, **Layer Normalization**, and **Feed-Forward Networks** refine data representations, resulting in precise anomaly classifications and threat analyses.

By leveraging their contextual and semantic capabilities, LLMs achieve so-

phisticated intrusion detection, excelling at recognizing and analyzing intrusion patterns.

1.7.3. A Review of LLM-Based IDS

Res-TranBiLSTM: Wang et al. [2023] combines CNN, LSTM, and transformers for IoT intrusion detection, with multi-head attention preceding BiLSTM to capture both local and global features. ResNet is used for spatial feature extraction on MQTT-based traffic data, providing efficient and accurate detection at the fog layer.

R1DIT: This privacy-preserving malware traffic classifier Barut et al. [2022] uses 2-D positional encoding and multi-head attention, focusing on TLS, DNS, and HTTP headers to anonymize data and improve classification performance without revealing sensitive information.

Flow-BERT-NIDS: Nguyen and Watabe [2022] uses BERT’s language modeling for flow-based NIDS, effectively managing domain shifts across different datasets.

CANBERT: Nwafor and Olufowobi [2022] detects malicious activities on in-vehicle CAN bus networks, identifying attacks such as DoS and impersonation using real-world data.

Bi-directional-GPT-IDS: Nam et al. [2021] is a GPT-based IDS for CAN bus systems, treating CAN ID sequences as sentences and using a bi-directional approach for efficient pattern recognition.

TAN-IDS: Nguyen et al. [2023] is a transformer attention network for in-vehicle CAN bus intrusion detection, outperforming RNN with self-attention and transfer learning.

F2I-ViT-IDS: Ho et al. [2022] is a vision transformer IDS that converts network flow data into images, achieving high accuracy in binary and multi-class classification.

BFS-NID: Du et al. [2023b] is a few-shot, class-incremental learning IDS that uses ViT for continuous learning of new attacks with minimal samples.

DCT-GAN: Li et al. [2021] is a GAN with a Dilated Convolutional Transformer that enhances anomaly detection by extracting fine-grained and coarse-grained features from time-series data.

CNN-Transformer-IDS: Yao et al. [2023] is a hybrid CNN-transformer IDS for Advanced Metering Infrastructure, integrating feature screening, sampling, and deep learning for robust classification.

KnowPhish: Li et al. [2024] is a multimodal LLM-based phishing detection system that builds a large-scale multimodal brand knowledge base, using LLMs to extract text from HTML and enhance phishing detection with or without logos.

These examples highlight the growing role of LLMs in IDS, demonstrating their adaptability in tackling modern cybersecurity challenges.

We summarize LLMs for intrusion detection and present them in Tab. 1.6.

1.7.4. Open Research Challenges

LLMs for intrusion detection present unique open research challenges:

- **Enhanced Context Handling:** Improving long-context management is essential, given current token limitations in LLMs. Future work could explore vector database solutions for retaining context over extended periods.
- **Minimizing Hallucinations:** LLMs may produce "hallucinations" or in-

Table 1.6: Review of LLMs for Intrusion Detection

Method	Key Approach	LLM Scheme
Res-TranBiLSTM (2023)	Combines CNN, BiLSTM, and transformers with attention to IoT detection.	Multi-Head-Attention
RIDIT (2022)	Classifies TLS, DNS, and HTTP headers using positional encoding and attention.	Multi-Head-Attention
Flow-BERT-NIDS (2022)	Leverages BERT for flow-based NIDS across varying datasets.	BERT
CANBERT (2022)	Detects CAN bus attacks with BERT-based modeling.	BERT
Bi-directional-GPT-IDS (2021)	Analyzes CAN ID sequences as sentences using GPT.	GPT
TAN-IDS (2023)	Uses self-attention for CAN bus intrusion detection.	Self-Attention
F2I-ViT-IDS (2022)	Converts flow data to images for ViT-based classification.	ViT
BFS-NID (2023)	Few-shot learning of new attacks using ViT.	ViT
DCT-GAN (2021)	Detect anomalies with a GAN-based transformer hybrid for time-series data.	GAN-Transformer
CNN-Transformer-IDS (2023)	Combines CNN and transformers for AMI intrusion detection.	CNN-Transformer
KnowPhish (2024)	Multimodal LLM detects phishing using text and logos.	Multimodal-LLM

accurate outputs, so ongoing research to reduce these occurrences through advanced prompt engineering and chain-of-thought reasoning is necessary for enhancing reliability.

- **Explainability Concerns:** The opacity of LLMs’ decision-making processes limits interpretability, which is critical for security applications. Integrating XAI techniques can help make LLM decisions more transparent and trustworthy.
- **Human-in-the-Loop:** Automation has limitations; therefore, integrating human expertise, such as reinforcement learning from human feedback (RLHF), through a feedback loop can help handle complex penetration testing tasks, improving the tool’s practical utility.
- **Complex Network Conditions:** Adapting LLMs to continuously evolving network conditions requires regular updates and retraining. Expanding multimodal knowledge sources is needed to improve the system’s adaptability.

- **LLM-Related Enhancements:** Techniques like retrieval augmentation can further enhance LLMs' detection capabilities by allowing access to real-time external knowledge, improving responsiveness to new phishing tactics, and enabling continuous knowledge updates.
- **Robustness Against Adversarial Attacks:** Developing defenses against adversarial example attacks Jin et al. [2024], such as prompt injection and text-to-image manipulation, is necessary to improve detection resilience.

1.8. Summary

ML enhances intrusion detection by enabling the detection of previously unseen **zero-day attacks** through the analysis of patterns in normal and abnormal behavior. This capability is crucial for defending against emerging threats that lack known signatures. Building on this cornerstone, a notable strength of ML-IDS is its ability to offer **explainability** using XAI, which provides insights into why certain activities are flagged as malicious. This enhances transparency, trustworthiness, and understanding for intrusion response. Additionally, ML-IDS excels at **identifying intrusions within encrypted traffic** by leveraging features beyond payload analysis, maintaining security even in encrypted environments. GNNs introduce **context-aware threat reasoning** to ML-IDS, modeling complex relationships and interactions among network entities. This approach enhances the system's capacity to detect sophisticated attacks, such as APTs, that involve multi-step or distributed components. Furthermore, the integration of advanced LLMs enables ML-IDS to process network data as language, providing a deeper **understanding of intrusions** and improving the recognition and analysis of threats.

In summary, ML extends traditional IDSs by integrating learning technologies that provide the five key capabilities discussed. As research advances, ML-driven innovations promise to further enhance IDSs, driving their continuous development and improvement.

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